

The cylindric Lorentzian problem at $\ell = 3$: a reduction to two combinatorial inequalities

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29 September 2026

Abstract

Let $s_{\lambda/\mu}^c$ be the cylindric skew Schur polynomial of a cylindric shape at level n with m beads, and let $N(s^c) = \sum_{\alpha} (K_{\alpha}^c / \alpha!) x^{\alpha}$ be its Brändén–Huh normalisation. The open regime of the conjecture “ $N(s^c)$ is Lorentzian” is $\ell \geq 3$ and $m \geq 2$. This note does not prove the conjecture at $\ell = 3$. It does three things. First, it *reduces* the $\ell = 3$ case to two explicit combinatorial inequalities on the coefficient array, by proving an elementary criterion (Lemma 3) for a nonnegative symmetric 3×3 matrix to have at most one positive eigenvalue: conditions (A) $M_{ij}^2 \geq M_{ii}M_{jj}$ and (B) $M_{ik}M_{kj} \geq M_{ij}M_{kk}$. Together these say that $\phi_{ij} = \log(M_{ij}/\sqrt{M_{ii}M_{jj}})$ is a pseudometric. Condition (B) is new; it is precisely what the previously recorded dead end `rlc-implies-13` was missing, and it is violated by that dead end’s own witness. Second, it establishes the *scope* of the criterion: Lemma 3 is strictly a 3×3 phenomenon, failing at $\ell = 4$ and, structurally, at $\ell = 5$ via the $K_{2,3}$ graph metric. Third, it proves the two inequalities in the totally ordered case (Lemmas 11 and 12) and isolates the remaining gap to a single precisely stated failure: monotone likelihood ratio holds across *comparable* inner shapes but genuinely fails across *incomparable* ones, while the fibrewise sums nevertheless satisfy it. Conditions (A) and (B) are verified with exact integer arithmetic on 220,658 Hessians.

1 Setting

1.1 Cylindric shapes and cylindric Schur polynomials

Fix integers $n > m \geq 1$. A *cylindric shape* at level n with m beads is a bi-infinite strictly increasing sequence $(x_i)_{i \in \mathbb{Z}}$ with $x_{i+m} = x_i + n$; we store it as the tuple $x = (x_1, \dots, x_m)$ with $x_1 < \dots < x_m < x_1 + n$, and we always read indices cyclically through $x_{i+m} = x_i + n$. Write $\nu \subseteq \rho$ for $x_i(\nu) \leq x_i(\rho)$ for all i , and $|\rho/\nu| = \sum_{i=1}^m (x_i(\rho) - x_i(\nu))$.

The pair ρ/ν is a *cylindric horizontal strip* when

$$\nu_i \leq \rho_i < \nu_{i+1} \quad (1 \leq i \leq m), \tag{1}$$

with $\nu_{m+1} = \nu_1 + n$. A *cylindric SSYT* of shape λ/μ with ℓ rows is a chain

$$\mu = \kappa^0 \subseteq \kappa^1 \subseteq \dots \subseteq \kappa^{\ell} = \lambda$$

in which every κ^t/κ^{t-1} is a cylindric horizontal strip; its weight is $\alpha = (\alpha_1, \dots, \alpha_{\ell})$ with $\alpha_t = |\kappa^t/\kappa^{t-1}|$. Set

$$K_{\alpha}^c = \#\{\text{cylindric SSYT of shape } \lambda/\mu \text{ and weight } \alpha\}, \quad s_{\lambda/\mu}^c(x_1, \dots, x_{\ell}) = \sum_{\alpha} K_{\alpha}^c x^{\alpha}.$$

This is homogeneous of degree $d = |\lambda/\mu|$ and is a symmetric polynomial; in particular $K_{\alpha}^c = K_{w\alpha}^c$ for every permutation w of the ℓ coordinates. That symmetry is used repeatedly below and is not re-proved here.

1.2 The normalisation and the three conditions

Put $N(s^c) = \sum_{\alpha} \frac{K_{\alpha}^c}{\alpha!} x^{\alpha}$. The conditions defining Lorentzian-ness in the form used throughout this programme are

- (L1) all coefficients are nonnegative — immediate, since K_{α}^c counts tableaux;
- (L2) $\text{supp}(s^c)$ is M-convex — already **proved** and Lean-verified in this programme (registry `cylindric-M-convex` `SortedBridge.sorted_exchange`, sorry-free);
- (L3) for every $\beta \in \mathbb{N}^{\ell}$ with $|\beta| = d - 2$, the quadratic form $\partial^{\beta} N(s^c)$ has at most one positive eigenvalue.

By the recorded **raw-hessian-lemma** (**proved**, and re-checked symbolically in this session), the matrix of the quadratic form $\partial^{\beta} N(s^c)$ is

$$M_{ij}^{(\beta)} = K_{\beta+e_i+e_j}^c, \quad (2)$$

the *raw* coefficients, not the normalised ones. All matrices below are the raw ones.

Throughout, $M = M^{(\beta)}$ is symmetric with nonnegative integer entries, and $\text{tr } M \geq 0$.

2 The two conditions and the criterion

Definition 1. Let M be a symmetric $\ell \times \ell$ matrix with nonnegative entries. Say M satisfies

$$M_{ij}^2 \geq M_{ii}M_{jj} \quad \text{for all } i \neq j, \quad (\text{A})$$

$$M_{ik}M_{kj} \geq M_{ij}M_{kk} \quad \text{for all distinct } i, j, k. \quad (\text{B})$$

Remark 2 (Why these two). If every $M_{ii} > 0$, set $N_{ij} = M_{ij}/\sqrt{M_{ii}M_{jj}}$, so that N is congruent to M and $N_{ii} = 1$. Then (A) says $N_{ij} \geq 1$ and (B) says $N_{ij} \leq N_{ik}N_{kj}$. Writing $\phi_{ij} = \log N_{ij}$, the pair (A)+(B) says exactly that ϕ is a *pseudometric*: nonnegative, symmetric, vanishing on the diagonal, and satisfying the triangle inequality. This is the shortest description of the criterion, and it is the one that explains its scope (§2.2). The proof below is written without dividing by diagonal entries, so that the degenerate cases $M_{ii} = 0$ need no separate treatment.

In terms of the coefficients, with $\alpha = \beta + e_i + e_j$, condition (A) is

$$(K_{\alpha}^c)^2 \geq K_{\alpha+e_i-e_j}^c K_{\alpha-e_i+e_j}^c,$$

which is exactly the raw root-log-concavity (RLC) already instrumented in this programme. Condition (B) is new.

Lemma 3 (Metric criterion). *Let M be a symmetric 3×3 matrix with nonnegative real entries satisfying (A) and (B). Then $e_2(M) \leq 0$, $\det M \geq 0$, and M has at most one positive eigenvalue. Moreover $\det M = 0$ if and only if at least two of the three inequalities (B1)–(B3) below are equalities.*

Proof. Write

$$P = M_{11}M_{22}M_{33}, \quad t = M_{12}M_{13}M_{23}, \quad x = M_{11}M_{23}^2, \quad y = M_{22}M_{13}^2, \quad z = M_{33}M_{12}^2.$$

Expanding the determinant of a symmetric 3×3 matrix gives the identity

$$\det M = P + 2t - x - y - z, \quad (3)$$

and directly from the definitions,

$$xyz = Pt^2. \quad (4)$$

Step 1: (B) gives $x, y, z \leq t$. Taking (B) with $k = 1$, $(i, j) = (2, 3)$ reads $M_{21}M_{13} \geq M_{23}M_{11}$; multiplying both sides by $M_{23} \geq 0$,

$$t = M_{12}M_{13}M_{23} \geq M_{11}M_{23}^2 = x. \quad (B1)$$

Taking $k = 2$, $(i, j) = (1, 3)$ reads $M_{12}M_{23} \geq M_{13}M_{22}$; multiplying by $M_{13} \geq 0$,

$$t \geq M_{22}M_{13}^2 = y. \quad (B2)$$

Taking $k = 3$, $(i, j) = (1, 2)$ reads $M_{13}M_{32} \geq M_{12}M_{33}$; multiplying by $M_{12} \geq 0$,

$$t \geq M_{33}M_{12}^2 = z. \quad (B3)$$

Step 2: $\det M \geq 0$. Suppose first $t = 0$. Then (B1)–(B3) force $x = y = z = 0$, and (A) multiplied out gives $t^2 = M_{12}^2M_{13}^2M_{23}^2 \geq (M_{11}M_{22})(M_{11}M_{33})(M_{22}M_{33}) = P^2$, so $P = 0$. By (3), $\det M = 0$.

Now suppose $t > 0$ and put $\xi = x/t$, $\eta = y/t$, $\zeta = z/t$, which lie in $[0, 1]$ by Step 1 and nonnegativity. By (4), $P = xyz/t^2 = t\xi\eta\zeta$, so (3) becomes

$$\det M = t(\xi\eta\zeta - \xi - \eta - \zeta + 2) = tg(\xi, \eta, \zeta), \quad g(\xi, \eta, \zeta) := \xi\eta\zeta - \xi - \eta - \zeta + 2.$$

The function g is affine in each variable separately, hence attains its minimum on the cube $[0, 1]^3$ at a vertex. At a vertex with r coordinates equal to 1 its value is

$$g = \begin{cases} 2, & r = 0, \\ 1, & r = 1, \\ 0, & r = 2, \\ 0, & r = 3, \end{cases}$$

so $g \geq 0$ on $[0, 1]^3$, and $\det M \geq 0$. Since g is affine in each variable and strictly positive at every vertex with $r \leq 1$, g vanishes exactly on the set where at least two of ξ, η, ζ equal 1; this is the stated equality condition.

Step 3: $e_2(M) \leq 0$. The second elementary symmetric function of the eigenvalues equals the sum of the three 2×2 principal minors,

$$e_2(M) = \sum_{i < j} (M_{ii}M_{jj} - M_{ij}^2),$$

and each summand is ≤ 0 by (A).

Step 4: conclusion. Let $\lambda_1 \geq \lambda_2 \geq \lambda_3$ be the eigenvalues and suppose $\lambda_2 > 0$. Then $\lambda_1 \geq \lambda_2 > 0$, and $\det M = \lambda_1\lambda_2\lambda_3 \geq 0$ forces $\lambda_3 \geq 0$. Hence $e_2(M) = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 \geq \lambda_1\lambda_2 > 0$, contradicting Step 3. So $\lambda_2 \leq 0$. $\square$

Remark 4. Steps 1–2 use only (B) (with (A) needed solely in the degenerate case $t = 0$), and Step 3 uses only (A). The roles are cleanly separated: (A) controls e_2 , (B) controls the determinant.

The converse direction of Step 4 also holds and is worth recording, because it says that at $\ell = 3$ nothing is lost by the reformulation.

Proposition 5 (Reduction). *Let M be a symmetric 3×3 matrix with nonnegative entries, so that $\text{tr } M \geq 0$. Then M has at most one positive eigenvalue if and only if $e_2(M) \leq 0$ and $\det M \geq 0$.*

Proof. $(\Leftarrow)$ is Step 4 above. $(\Rightarrow)$ Let $\lambda_1 \geq 0 \geq \lambda_2 \geq \lambda_3$. Then $\det M = \lambda_1 \lambda_2 \lambda_3 \geq 0$. Moreover $\text{tr } M \geq 0$ gives $\lambda_1 \geq -(\lambda_2 + \lambda_3) \geq 0$, and since $\lambda_2 + \lambda_3 \leq 0$,

$$e_2(M) = \lambda_1(\lambda_2 + \lambda_3) + \lambda_2 \lambda_3 \leq -(\lambda_2 + \lambda_3)^2 + \lambda_2 \lambda_3 = -(\lambda_2^2 + \lambda_2 \lambda_3 + \lambda_3^2) \leq 0. \quad \square$$

Corollary 6. *At $\ell = 3$, condition (L3) for a given β is equivalent to the conjunction of RLC at β and $\det M^{(\beta)} \geq 0$. Since RLC has been verified without exception throughout this programme, **the entire open content of (L3) at $\ell = 3$ is the single determinantal inequality $\det M^{(\beta)} \geq 0$.***

2.1 Condition (B) is exactly what the recorded dead end was missing

The registry node `rlc-implies-13` is a **dead-end** whose witness is

$$W = \begin{pmatrix} 4 & 6 & 4 \\ 6 & 1 & 4 \\ 4 & 4 & 4 \end{pmatrix},$$

a strictly positive symmetric matrix satisfying every 2×2 condition ($36 \geq 4$, $16 \geq 16$, $16 \geq 4$) yet having characteristic polynomial $x^3 - 9x^2 - 44x + 16$ and therefore two positive eigenvalues. In the present language: W satisfies (A) and $\det W = -16 < 0$. Checking (B) at $(i, j, k) = (1, 2, 3)$: $W_{13}W_{32} = 16$ while $W_{12}W_{33} = 24$, so (B) *fails*. Equivalently, in the normalisation of Remark 2, $(N_{12}, N_{13}, N_{23}) = (3, 1, 2)$ and the triangle inequality $N_{12} \leq N_{13}N_{23} = 2$ is violated. Condition (B) is thus not an arbitrary strengthening: it is precisely the hypothesis whose absence produced the recorded counterexample.

2.2 Scope of Lemma 3

It matters, for this programme, to record not only that a reason is true but for which quantifiers. Lemma 3 is *strictly a 3×3 statement*.

Proposition 7. *For $\ell \geq 4$ there exist symmetric $\ell \times \ell$ matrices with nonnegative entries satisfying (A) and (B) and having two positive eigenvalues.*

Proof and structural witness. A random integer search at $\ell = 4$ found 777 such matrices among 4012 satisfying (A) and (B), for example $\begin{pmatrix} 3 & 6 & 2 & 6 \\ 6 & 0 & 1 & 4 \\ 2 & 1 & 0 & 4 \\ 6 & 4 & 4 & 1 \end{pmatrix}$. The conceptual reason appears at $\ell = 5$.

Suppose all $M_{ii} > 0$ and normalise as in Remark 2, so $N_{ij} = e^{\phi_{ij}}$ with ϕ a pseudometric. On the hyperplane $\{\sum_i x_i = 0\}$ one computes, using $N_{ii} = 1$,

$$x^\top N x = \sum_i x_i^2 + \sum_{i \neq j} e^{\phi_{ij}} x_i x_j = \left(\sum_i x_i \right)^2 + \sum_{i \neq j} (e^{\phi_{ij}} - 1) x_i x_j = \sum_{i \neq j} (e^{\phi_{ij}} - 1) x_i x_j.$$

So N has at most one positive eigenvalue as soon as the nonnegative hollow matrix $(e^{\phi_{ij}} - 1)$ is conditionally negative definite, i.e. “of negative type”. Not every metric is of negative type, and the smallest classical obstruction is the shortest-path metric of $K_{2,3}$. Taking ϕ to be that metric on five points ($\phi = 1$ across the parts, $\phi = 2$ within each part) and $N_{ij} = e^{\phi_{ij}}$, conditions (A) and (B) hold, while the spectrum is $\{-6.389, -6.389, -6.389, 4.469, 19.698\}$: two positive eigenvalues. An exact integer rescaling confirms this with rational arithmetic. $\square$

Proposition 7 does *not* refute (L3) for cylindric shapes at $\ell \geq 4$; the cylindric pseudometrics may all be of negative type. It says only that the criterion used here cannot be the mechanism beyond $\ell = 3$. This is exactly why $\ell = 3$ was the right target, and the reason is now explained rather than merely chosen.

3 The two inequalities as combinatorics

Because s^c is symmetric, the three instances of (A) are equivalent to one another, and likewise the three instances of (B). Writing $c = d - a - b$ and

$$k(a, b) := K_{(a, b, d-a-b)}^c,$$

the six Hessian inequalities collapse to two statements about the triangular array k .

Proposition 8. *At $\ell = 3$, condition (A) holds for all β if and only if k is log-concave in a ,*

$$k(a, b)^2 \geq k(a+1, b)k(a-1, b),$$

and condition (B) holds for all β if and only if k is log-submodular,

$$k(a+1, b)k(a, b+1) \geq k(a, b)k(a+1, b+1).$$

Verification. Place $\beta = (p, q, r)$ and let $A = \beta + 2e_1$, $B = \beta + 2e_2$, $C = \beta + 2e_3$, $F = \beta + e_1 + e_2$, $G = \beta + e_1 + e_3$, $H = \beta + e_2 + e_3$, so that $M = \begin{pmatrix} K_A & K_F & K_G \\ K_F & K_B & K_H \\ K_G & K_H & K_C \end{pmatrix}$. Then (A) reads $K_F^2 \geq K_A K_B$, $K_G^2 \geq K_A K_C$, $K_H^2 \geq K_B K_C$, and (B) reads $K_G K_H \geq K_F K_C$, $K_F K_H \geq K_G K_B$, $K_F K_G \geq K_H K_A$. In (a, b) coordinates, $A = (p+2, q)$, $B = (p, q+2)$, $C = (p, q)$, $F = (p+1, q+1)$, $G = (p+1, q)$, $H = (p, q+1)$; substituting turns (A)₁₃ into log-concavity in a and (B) _{$k=3$} into log-submodularity, and the S_3 -symmetry of K^c carries each family onto the displayed representative. The equivalence was additionally checked mechanically on all 573 instances with $n \leq 6$, $d \leq 9$: zero mismatches in either direction. $\square$

4 An interlacing model

The following description of K^c at $\ell = 3$ is the foundation for the partial results of §5. It removes the chain-of-shapes formulation in favour of a single cyclically interlacing sequence with box constraints.

Proposition 9. *Fix a cylindric skew shape λ/μ at level n with m beads. Then*

$$k(a, b) = \# \left\{ (\kappa, \sigma) \in \mathbb{Z}^m \times \mathbb{Z}^m : \begin{array}{ll} \mu_i \leq \kappa_i \leq \mu_{i+1} - 1, & \lambda_{i-1} + 1 \leq \sigma_i \leq \lambda_i, \\ \kappa_i \leq \sigma_i \leq \kappa_{i+1} - 1 & (1 \leq i \leq m), \\ S_\kappa = |\mu| + a, & S_\sigma = |\mu| + a + b \end{array} \right\},$$

where $S_\kappa = \sum_i \kappa_i$, indices are read cyclically through $x_{i+m} = x_i + n$, and $|\mu| = \sum_i \mu_i$.

Proof. A cylindric SSYT with 3 rows is a chain $\mu \subseteq \kappa \subseteq \sigma \subseteq \lambda$ of horizontal strips. By (1), κ/μ is a strip iff $\mu_i \leq \kappa_i < \mu_{i+1}$; σ/κ is a strip iff $\kappa_i \leq \sigma_i < \kappa_{i+1}$; and λ/σ is a strip iff $\sigma_i \leq \lambda_i$ and $\lambda_i < \sigma_{i+1}$, i.e. $\lambda_{i-1} + 1 \leq \sigma_i \leq \lambda_i$. No separate check that κ and σ are shapes is needed: $\kappa_i \leq \mu_{i+1} - 1 < \mu_{i+1} \leq \kappa_{i+1}$ gives $\kappa_i < \kappa_{i+1}$, and likewise for σ . The weights are $a = |\kappa/\mu| = S_\kappa - |\mu|$, $b = |\sigma/\kappa| = S_\sigma - S_\kappa$, $c = |\lambda/\sigma| = |\lambda| - S_\sigma$. $\square$

Proposition 9 was checked against the independent transfer-matrix dynamic program on all 911 instances with $n \leq 6$, $d \leq 8$: zero disagreements.

Eliminating κ from the middle constraint exhibits (κ, σ) as a single cyclic chain

$$\kappa_1 \leq \sigma_1 < \kappa_2 \leq \sigma_2 < \cdots \leq \sigma_m < \kappa_1 + n,$$

so the constraint system consists solely of differences $x_p - x_q \leq c$. In particular the solution set $\mathcal{L} \subseteq \mathbb{Z}^{2m}$ is closed under coordinatewise min and max, i.e. it is a sublattice of $\mathbb{Z}^{2m}$, and (S_κ, S_σ) is a linear — hence modular — functional on it. Writing $P(u, v) = \#\{(\kappa, \sigma) \in \mathcal{L} : S_\kappa = u, S_\sigma = v\}$, so that $k(a, b) = P(|\mu| + a, |\mu| + a + b)$, Proposition 8 says that (B) is exactly the statement that P is *log-supermodular* on $\mathbb{Z}^2$.

Remark 10 (Why Ahlswede–Daykin does not apply directly). Log-supermodularity of a pushforward follows from the four functions theorem when the projection is a *lattice homomorphism*. Here it is not: for $x, y \in \mathcal{L}$, $S_{\kappa^x \vee \kappa^y} = \sum_i \max(\kappa_i^x, \kappa_i^y) \geq \max(S_{\kappa^x}, S_{\kappa^y})$, and the inequality is generally strict. Modularity gives $\pi(x \wedge y) + \pi(x \vee y) = \pi(x) + \pi(y)$ with $\pi(x \vee y) \geq \pi(x) \vee \pi(y)$, so the pair $(x \wedge y, x \vee y)$ lands *outside* the fibres $\pi(x) \wedge \pi(y)$ and $\pi(x) \vee \pi(y)$ by a nonnegative slack. This obstruction is real and is the reason the argument of §5 takes a different route.

5 Partial results: the totally ordered case

Fix κ admissible. By Proposition 9, once κ is fixed each σ_i ranges *independently* over the interval $[L_i(\kappa), R_i(\kappa)]$ with

$$L_i(\kappa) = \max(\kappa_i, \lambda_{i-1} + 1), \quad R_i(\kappa) = \min(\kappa_{i+1} - 1, \lambda_i).$$

Hence

$$P(u, v) = \sum_{\kappa : S_\kappa = u} W_\kappa(v), \quad W_\kappa = \mathbf{1}_{[L_1, R_1]} * \cdots * \mathbf{1}_{[L_m, R_m]}. \quad (5)$$

Each W_κ is a convolution of interval indicators, hence a Pólya frequency sequence of order 2 (PF₂): log-concave with no internal zeros.

Recall that $f \preceq_{\text{lr}} g$ (the *likelihood ratio* order, equivalently TP₂ of the two-row matrix) means $f(v)g(v') \geq f(v')g(v)$ for all $v < v'$; and that a nonnegative sequence g is PF₂ iff

$$g(p)g(p' + \delta) \geq g(p + \delta)g(p') \quad \text{for all } p > p', \delta \geq 0. \quad (6)$$

Lemma 11 (Convolution preserves the likelihood ratio order). *Let f, f', g, g' be nonnegative PF₂ sequences with $f \preceq_{\text{lr}} f'$ and $g \preceq_{\text{lr}} g'$. Then $f * g \preceq_{\text{lr}} f' * g'$.*

Proof. By transitivity it suffices to prove $f * g \preceq_{\text{lr}} f' * g$ for $f \preceq_{\text{lr}} f'$ and g PF₂ (the second step, $f' * g \preceq_{\text{lr}} f' * g'$, is the same statement with the roles of the factors exchanged). Fix $v < v'$ and set $\delta = v' - v > 0$. Then

$$S := (f * g)(v)(f' * g)(v') - (f * g)(v')(f' * g)(v) = \sum_{w, w'} f(w)f'(w') \Delta(w, w'),$$

where $\Delta(w, w') = g(v - w)g(v' - w') - g(v' - w)g(v - w')$. Exchanging the summation names $w \leftrightarrow w'$ and using $\Delta(w', w) = -\Delta(w, w')$ gives $S = -\sum_{w, w'} f(w')f'(w)\Delta(w, w')$, whence

$$2S = \sum_{w, w'} [f(w)f'(w') - f(w')f'(w)] \Delta(w, w').$$

Both brackets are antisymmetric in (w, w') . For $w < w'$ the first is ≥ 0 because $f \preceq_{\text{lr}} f'$. For the second, put $p = v - w$ and $p' = v - w'$, so $p > p'$; then $v' - w = p + \delta$ and $v' - w' = p' + \delta$, and

$$\Delta(w, w') = g(p)g(p' + \delta) - g(p + \delta)g(p') \geq 0$$

by (6). So for $w < w'$ both factors are ≥ 0 , for $w > w'$ both are ≤ 0 , and for $w = w'$ the product vanishes. Every term of $2S$ is therefore ≥ 0 , so $S \geq 0$. $\square$

Lemma 12 (Comparable inner shapes). *If $\kappa \leq \kappa'$ coordinatewise and both are admissible, then $W_\kappa \preceq_{\text{lr}} W_{\kappa'}$.*

Proof. L_i is nondecreasing in κ_i and R_i is nondecreasing in κ_{i+1} , so $L_i(\kappa) \leq L_i(\kappa')$ and $R_i(\kappa) \leq R_i(\kappa')$ for every i . For two intervals with both endpoints weakly increasing, $\mathbf{1}_{[L,R]} \preceq_{\text{lr}} \mathbf{1}_{[L',R']}$: if $v < v'$ and the right-hand side of the defining inequality is nonzero, then $L' \leq v < v' \leq R$, whence $v \geq L' \geq L$ and $v < v' \leq R$ give $v \in [L, R]$, and $v' > v \geq L'$ with $v' \leq R \leq R'$ give $v' \in [L', R']$, so the left-hand side is 1 as well. Interval indicators are PF_2 and convolutions of PF_2 sequences are PF_2 , so Lemma 11 applies inductively over $i = 1, \dots, m$. $\square$

Corollary 13. *Condition (B), hence (L3) together with RLC, holds at $\ell = 3$ for every level n whenever the set of admissible inner shapes κ is a chain under containment — in particular for $m = 1$.*

Proof. If the admissible κ form a chain graded by S_κ , then for $u < u'$ every κ in the fibre $\{S_\kappa = u\}$ is $\leq$ every κ' in $\{S_{\kappa'} = u'\}$. Summing a TP_2 matrix over consecutive blocks of rows preserves TP_2 , since each 2×2 minor of the blocked matrix is a sum of 2×2 minors of the original with rows taken from the earlier and the later block respectively. Hence $P(u, \cdot) \preceq_{\text{lr}} P(u', \cdot)$ by (5) and Lemma 12, which is (B). At $m = 1$ the shapes κ are single integers, so the hypothesis holds. $\square$

6 The gap, stated precisely

Corollary 13 is exactly as far as the totally ordered argument reaches, and the obstruction to going further is not a missing estimate but a false statement. The natural strengthening — “ $S_\kappa < S_{\kappa'}$ implies $W_\kappa \preceq_{\text{lr}} W_{\kappa'}$ ” — is **false**.

Proposition 14. *There exist admissible κ, κ' with $S_\kappa < S_{\kappa'}$ and $W_\kappa \not\preceq_{\text{lr}} W_{\kappa'}$. For instance at $n = 5$, $m = 2$, $\mu = (0, 2)$, $\lambda = (1, 5)$, take $\kappa = (1, 2)$ and $\kappa' = (0, 4)$, so $S_\kappa = 3 < 4 = S_{\kappa'}$. Then W_κ is supported on $\{3, 4, 5, 6\}$ with all values 1, while $W_{\kappa'}$ is supported on $\{5\}$ with value 1. At $(v, v') = (5, 6)$,*

$$W_\kappa(5) W_{\kappa'}(6) = 1 \cdot 0 = 0 < 1 = 1 \cdot 1 = W_\kappa(6) W_{\kappa'}(5).$$

Over the range $n \leq 7$, $m \geq 2$, $d \leq 9$, the comparable case of Lemma 12 holds on all 15,147 pairs tested, while the incomparable case fails on 692 of 1,652 pairs — a 42% failure rate, not a rare degeneracy. And yet the fibrewise sums satisfy the order on all 327 pairs tested. This localises the gap exactly:

Gap. The likelihood ratio order holds between W_κ and $W_{\kappa'}$ when $\kappa \leq \kappa'$, fails in general when $S_\kappa < S_{\kappa'}$ but κ, κ' are incomparable, and is nevertheless restored after summing each fibre $\{S_\kappa = u\}$. The missing step is therefore a *cancellation across an antichain*, not a pointwise comparison. No argument that compares individual inner shapes can close it.

Condition (A) is not addressed by this route at all and remains open in the same regime.

7 Computational verification

All arithmetic is exact. Eigenvalue counts use the integer characteristic polynomial computed by Faddeev–LeVerrier over $\mathbb{Q}$ together with Descartes’ rule, which is exact for real-rooted polynomials; no floating point eigenvalues are used for any decision. Controls were re-run this session before the checker was aimed at anything new.

Control / test	instances	failures
Positive control: $N(s_\lambda)$ Lorentzian (untuned)	16	0
Negative control: detector must reject	3	3 rejected
Raw-Hessian lemma (2), symbolic	—	confirmed
Product-form (Theorem A) control	1105	0
Product-form negative control (f_t not log-concave)	5573	3604 fail (L3)
(A) and (B), $\ell = 3$, $n \leq 8$, $d \leq 12$	77283 Hessians	0, 0
(A) and (B), $\ell = 4, 5, 6$, $n \leq 6$, $d \leq 9$	143375 Hessians	0, 0
(L3) itself over the same ranges	220658 Hessians	0
Lemma 3, adversarial integer search	45301	0
negative control, (A) alone	54068	21135 have > 1 positive eig.
negative control, (B) alone	23089	526 have > 1 positive eig.
Equality criterion of Lemma 3	3921 Hessians	0 mismatches
Interlacing model vs. transfer-matrix DP	911	0 disagreements
Reformulation of Prop. 8	573	0 mismatches
Lemma 11, random PF_2 quadruples	97289	0
negative control, PF_2 dropped	484	99 fail
Lemma 12, comparable $\kappa \leq \kappa'$	15147 pairs	0
Incomparable κ , $S_\kappa < S_{\kappa'}$	1652 pairs	692 fail
Fibre sums F_u (this is (B))	327 pairs	0

A note on one figure. Of the 3921 Hessians in the $n \leq 6$, $d \leq 9$, $\ell = 3$ range, 2573 (65.6%) have $\det M = 0$. An earlier guess that this must be caused by a vanishing diagonal entry was *wrong*: 257 of them have all $M_{ii} > 0$, the all-ones matrix among them. The correct characterisation is the one delivered by the multilinear form of the proof — at least two of (B1)–(B3) are equalities — and it matches all 3921 cases.

8 Routes killed, with their reasons and the scope of each reason

- **Summand-wise arguments.** Lorentzian polynomials are not closed under addition: $(x + y)^2 + (2x + y)^2$ has a rank-2 positive semidefinite Hessian. *Scope: none* — this reason carries no hypothesis and kills every route that splits s^c into a sum and argues termwise. It is what shuts the Schur route even inside Postnikov’s positive range, where the negative-coefficient obstruction does not apply.
- **RLC alone.** Registry `rlc-implies-l3`, witness $\begin{pmatrix} 4 & 6 & 4 \\ 6 & 1 & 4 \\ 4 & 4 & 4 \end{pmatrix}$. *Scope: $\ell \geq 3$* , and the witness is 3×3 , so it lives in exactly today’s regime. This note identifies the missing hypothesis as (B).
- **The metric criterion beyond $\ell = 3$.** Proposition 7. *Scope: $\ell \geq 4$* . The criterion is a 3×3 phenomenon because it is equivalent to the negative-type property of $e^\phi - 1$, which no longer

follows from the triangle inequality once there are enough points; $K_{2,3}$ is the obstruction. This does not touch the truth of (L3) at $\ell \geq 4$.

- **Ahlswe–Daykin on the interlacing lattice.** (S_κ, S_σ) is modular but not a lattice homomorphism, so the four functions theorem gives no pointwise fibre inequality. *Scope: this particular projection*; a different grading of $\mathcal{L}$ is not excluded.
- **Ordering inner shapes by S_κ .** Proposition 14, 42% failure rate. *Scope: $m \geq 2$ with incomparable admissible κ* ; at $m = 1$ the order is total and Corollary 13 goes through.
- **Brändén–Huh closure theorems.** Not used anywhere in this note. The entry for [arXiv:1902.03719](#) is graded **agent-summary**, and reading it at source remains the highest-value next step for this programme; nothing has been smuggled in as well known.

9 What would close it

Two statements, both about the array $k(a, b) = K_{(a, b, d-a-b)}^c$, would together prove (L3) at $\ell = 3$ for every m , every n and every shape, by Lemma 3 and Proposition 8:

Conjecture 15. $k(a, b)^2 \geq k(a+1, b) k(a-1, b)$ for all a, b .

Conjecture 16. $k(a+1, b) k(a, b+1) \geq k(a, b) k(a+1, b+1)$ for all a, b ; equivalently, the fibre count $P(u, v)$ of the cyclic interlacing lattice $\mathcal{L}$ is log-supermodular on $\mathbb{Z}^2$.

Conjecture 16 holds when the admissible inner shapes form a chain (Corollary 13). The evidence for both is 220,658 Hessians without an exception, and the data further indicate that both hold at every ℓ tested, not only $\ell = 3$ — so whatever mechanism produces them is not a 3×3 accident, even though the criterion that consumes them is.